Estimating the Carrying Capacity for Black Rhinoceros (*Diceros bicornis*) in Selati Game Reserve

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Rationale

This report was prepared for the management of Selati Game Reserve to provide an evidence-based estimate of the ecological carrying capacity for black rhinos (*Diceros bicornis*). It synthesises available spatial data, habitat suitability modelling, and relevant ecological literature to inform conservation planning and long-term population management.

Summary

Estimating carrying capacity is essential for managing black rhinoceros populations within fenced reserves, where habitat quality and spatial constraints directly influence population size. By integrating key environmental variables such as vegetation, water availability, terrain, slope, and proximity to roads into a Habitat Suitability Index (HSI), spatially explicit habitat patches can be identified and quantified. Using GPS data and published home range sizes adjusted for overlap, carrying capacity can be estimated as the ratio of suitable habitat area to effective home range size. This approach revealed that over 80% of the area of Selati Game Reserve comprises suitable habitat, supporting an estimated population of approximately 40 (33 – 50) black rhinos, with home ranges closely aligned to good-quality habitat and water sources.

Introduction

Understanding the carrying capacity of enclosed areas is critical for the effective conservation of large herbivores such as black rhinoceros (*Diceros bicornis*). Carrying capacity represents the maximum number of individuals an environment can support indefinitely without causing long-term ecological degradation (Sinclair, Fryxell & Caughley, 2006). In the context of endangered megafauna, accurate estimates of carrying capacity are essential for guiding reintroductions, translocations, and population management strategies (Hayward & Kerley, 2009). For species with large spatial requirements and selective habitat preferences like the black rhino, such estimates depend on integrating habitat quality, resource availability, and spatial ecology.

Quantifying carrying capacity involves evaluating key environmental variables such as vegetation type, water access, terrain, and human infrastructure, often through spatial modelling approaches. Geographic Information Systems (GIS) and remotely sensed data provide powerful tools to assess habitat suitability and translate it into potential population thresholds (Mills et al., 2001; Laver & Kelly, 2008). This is particularly relevant in fenced reserves where population growth may be limited by area rather than ecological interactions. This report estimates the carrying capacity for black rhinoceros (*Diceros bicornis*) within Selati Game Reserve (SeGR), South Africa, using a spatially explicit, habitat-based modelling approach.

Methods

SeGR is a private protected area located in Limpopo Province, South Africa, that is characterized by mixed woodland and shrubland (i.e. bushveld). The reserve covers approximately 30,000 hectares and exhibits varied topography. The climate is subtropical with distinct wet and dry seasons, influencing vegetation phenology and water availability. Dominant vegetation types include a mosaic of woodlands largely dominated by *Combretum* sp. and *Colophospermum mopane*, as well as other woody shrublands, and riverine forest patches. The reserve is bounded by a perimeter fence, creating a semi-closed system for wildlife management. Infrastructure such as roads

and water points are distributed across the landscape, influencing animal movement and habitat use.

A Habitat Suitability Index (HSI) was calculated using a weighted linear combination of spatial environmental variables known to influence black rhino habitat preferences. These variables included vegetation type, proximity to water sources, terrain ruggedness, slope, and distance from roads. Each variable was normalised to a 0–1 scale, and weights were assigned based on ecological relevance as advised by reserve staff and informed by the published literature (Jachmann, 2008; Okita-Ouma *et al.*, 2009; Young *et al.*, 2009; Ferreira *et al.*, 2015; Adcock & Fynn, 2020): vegetation suitability (classified from habitat categories) = 0.70; proximity to water (normalised inverse distance) = 0.15; fragmentation = 1; ruggedness and slope (flatter areas preferred) = 0.04 each; distance from roads (further preferred) = 0.02. All spatial layers were resampled to a common resolution and masked to the fenced boundary of the reserve to ensure spatial alignment.

A threshold (HSI > 0.5) was used to classify raster cells as suitable or unsuitable. Contiguous suitable cells were grouped into discrete patches using an 8-cell neighbourhood approach. Each patch was converted to a polygon and its area calculated in hectares using projected coordinates (UTM Zone 35S), appropriate for accurate areal measurements in Limpopo.

GPS and sighting data for known individual black rhinos were used to estimate minimum convex polygon (MCP) home ranges, along with data from the published literature (Conway & Goodman, 1989; Harris & Craig, 1997; Whyte & Joubert, 1998; Schrader & Ferreira, 2008; Plotz *et al.*, 2016). Home range sizes were calculated separately for males and females, and a weighted mean home range was derived based on an approximation of sex ratios (female:male ratio = 1.3:1) based on the published literature. Overlap between individual home ranges was incorporated using published values (20–50%; Odenhaal-Holmes *et al.,* 2014), resulting in effective home range areas (weighted mean home range * (1-overlap)) that were used to calculate density.

For each suitable habitat patch, carrying capacity was estimated as patch area (ha) divided by effective home range area (ha). Estimates were calculated

using minimum, maximum, and midpoint overlap assumptions, and rounded down to the nearest whole individual per patch. Patches supporting fewer than one individual were excluded.

Results

Black rhinos at SeGR currently exhibit home range sizes of between 922.45 ha (Msinyane, male) to 10,704 ha (Tshevu, male)(table 1). Average home range sizes were 5,029 ha (± 5,075 ha Standard Deviation) for males and 2,919 ha (± 1,982) for females. The small sample size and substantial variation rendered these data inappropriate for modelling using home range data from SeGR alone.

Table 1. Black rhinoceros ID, sex, and home range sizes derived from GPS and direct observational data in SeGR, South Africa.

		Home range size	
Rhino ID	**Sex**	**Area (ha)**	**Area (km^2)**
Epaki	M	3462.51	34.63
Iantha	F	3064.72	30.65
Leta	F	1417.67	14.18
Masinda	F	1525.99	15.26
Msinyane	M	922.45	9.22
Nandi	F	5671.04	56.71
Tshevu	M	10704.54	107.05

The habitat suitability index (HSI), integrating vegetation, terrain ruggedness, slope, proximity to water, and road distance, revealed spatial heterogeneity across the study area. Total suitable habitat area was estimated at 24,696.49 ha distributed among 2,914 discrete habitat patches, suggesting that 82.32 % of SeGR contains suitable habitat for black rhinoceros. The distribution of habitat patches indicates a mosaic landscape with variable quality and connectivity (fig. 1).

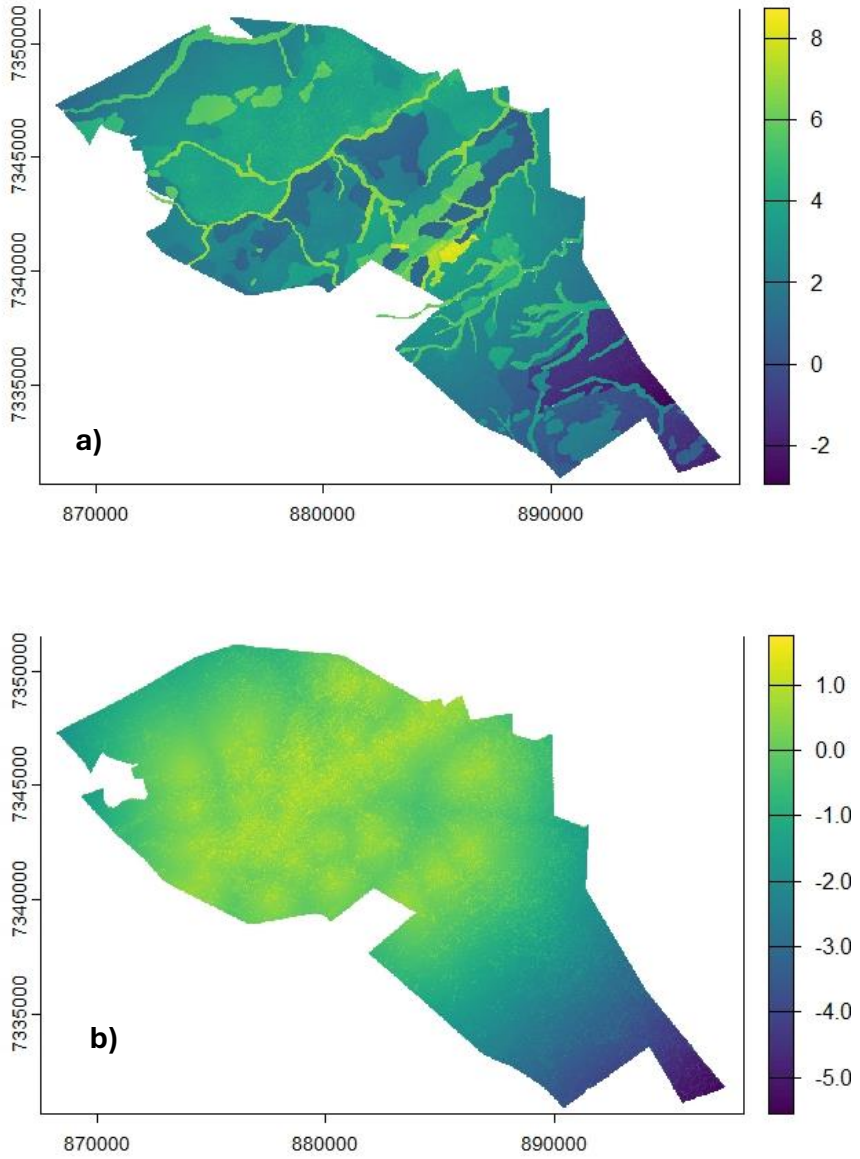

Figure 1. Habitat Suitability maps for black rhinoceros in SeGR. **a)** Including all variables; the map is dominated by vegetation due to the weight and geometry of the layer. **b)** Without vegetation, demonstrating the spatial variance and importance of all other variables.

Using sex-specific home range sizes and adjusted for home range overlap, the estimated carrying capacity of the reserve was calculated as 40 black rhinoceros individuals (95% confidence interval: 33 – 50). This estimate aligns with observed population densities reported in similar protected areas. Habitat patches contributing most to carrying capacity were typically larger and closer to critical resources such as

water and woody vegetation, underscoring the importance of these factors in black rhino spatial ecology.

Individual home ranges largely overlapped with areas of higher HSI, confirming the utility of the habitat suitability model in predicting key areas for black rhino activity. Known locations of individuals clustered near water sources and within high-quality habitat patches, reinforcing the significance of water availability and habitat composition in black rhino spatial distribution

Discussion

This study provides a spatially explicit estimate of black rhinoceros carrying capacity within SeGR, highlighting that a substantial portion of the reserve (over 80%) comprises suitable habitat, capable of supporting up to 50 individuals. The integration of relevant environmental variables into the Habitat Suitability Index (HSI) allowed for a detailed understanding of habitat quality and spatial heterogeneity. Patches near preferred vegetation types contributed most to carrying capacity. The use of GPS-derived home ranges and published overlap estimates further refined density calculations, aligning well with observed densities in similar reserves.

The only prior carrying capacity estimate for black rhinos in SeGR was calculated in 2010 (Goodman & Peel, 2010). That report used vegetation data at a coarser scale than that which was available for this report, hence could be considered less precise in that regard. Goodman & Peel (2010) also included precipitation and temperature in their calculations. These metrics are often excluded from HSI models due to their high temporal variability and the challenges of capturing their ecological effects in a stable, meaningful way (Austin, 2002; Guisan & Zimmermann, 2000). Both variables exhibit strong seasonal and interannual fluctuations, and their ecological consequences — such as changes in vegetation productivity or water availability — are often better represented through integrative or proximate variables like vegetation type, NDVI, or distance to water (Pettorelli et al., 2005). Including these variables without accounting for their variability and context can introduce noise and reduce the reliability of models, particularly in systems where species respond more to cumulative or indirect effects

than to raw climatic values. Goodman & Peel (2010) concluded that the ecological carrying capacity for SeGR was 63 individuals, 75% of which (i.e., 47 animals) was the highest density that could be achieved without density-dependent feedbacks. This estimate is in-line with the maximum value from the current data (i.e., 50 animals). Goodman & Peel (2010) also included competing browser biomass while that was omitted from this report. In systems where interspecific competition for forage is significant and particularly where resources are limited, omitting browser biomass can lead to overestimates of carrying capacity (Prins & van der Jeugd, 1993; Illius & O'Connor, 2000). However, Goodman & Peel (2010) estimated carrying capacity in SeGR by simply comparing its browser density to that of surrounding reserves, without accounting for critical ecological factors such as spatial variation in browse availability, topography, or the composition of the broader browser community. This comparative approach assumes that adjacent reserves are ecologically equivalent, which is rarely the case in heterogeneous savanna systems where vegetation structure, species composition, and water availability can vary markedly over short distances. Moreover, ignoring the influence of interspecific interactions and resource partitioning within the browser guild can lead to misleading conclusions, especially in areas where competition for forage or differences in feeding strategies (e.g., selective vs. generalist browsers) play a key role in shaping population dynamics (Du Toit & Cumming, 1999). Quantifying browser community structure and dynamics is beyond the remit of this report, particularly as it requires additional data that would incur substantial cost and survey effort. The model outputs presented in this report should therefore be taken as representing habitat potential rather than realised capacity. While the upper estimate of 50 animals corresponds with Goodman & Peel 's (2010) 75% estimate the above considerations still apply. Thus it would be reasonable to apply the precautionary principle and retain the modelled estimate of 40 black rhinos as the carrying capacity for SeGR, rather than adopting the upper confidence limit, to account for ecological uncertainty, potential competition, and long-term habitat sustainability.

An important consideration when interpreting carrying capacity estimates is the relationship between population density and home range size. In SeGR, observed home ranges for black rhinoceros vary widely, from approximately 900 ha to over 10,000 ha,

reflecting individual differences in space use likely influenced by resource availability, social dynamics, and population density. It is well documented that, range sizes can decrease with increased density due to increased competition and more efficient use of space, provided that resource availability remains adequate (Owen-Smith, 1988; Lent & Fike, 2003; Linklater *et al*., 2010; Odenkaal, 2012). These smaller, yet viable, home ranges can support more individuals within a fixed area without necessarily indicating overcrowding or ecological stress. Therefore, the estimated carrying capacity is not only plausible but also consistent with expected ecological patterns in closed systems. This reinforces the importance of monitoring how spatial behaviour and home range dynamics respond to changes in density over time.

Several limitations should be noted, however. First, the HSI relies on weighted environmental layers whose assigned weights, while informed by literature and expert opinion, may not capture all ecological nuances or temporal variation in habitat use. Secondly, the assumption of static home range sizes and overlap values may oversimplify complex individual movement behaviours, especially in a dynamic environment with seasonal resource fluctuations. Third, spatial resolution and classification threshold of habitat suitability could influence patch delineation and area estimates. Finally, as a fenced reserve, population dynamics may also be affected by factors such as social structure, poaching risk, and management interventions, which are not explicitly accounted for in the habitat-based carrying capacity model.

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